

HUMIDITY AND TEMPERATURE MONITORING

Using Arduino With The IoT

D. CITHARTHAN AND T. JONAH GILBERT

Using the Internet of Things (IoT) in homes and industries it is possible to control any electrical or electronic equipment. Moreover, you can get the information from any sensor and analyse it graphically or in any user-defined format from anywhere

in the world. The IoT using Arduino microcontroller (MCU) is easy and fun for those who are new to the field.

In this article, humidity and temperature information from DHT-11 sensor is analysed graphically on ThingSpeak platform using Arduino

MCU and ESP8266 Wi-Fi module. Block diagram of the whole setup is shown in Fig. 1.

Circuit and working

Circuit diagram for monitoring humidity and temperature is shown in Fig. 2. It is built around Arduino MCU, DHT11 sensor and ESP8266 Wi-Fi module.

The DHT11 sensor senses humidity and temperature, and sends the information to digital pin 5 of Arduino MCU, as shown in Fig. 2. From Arduino MCU, humidity and temperature values are uploaded to the Cloud at regular intervals of time through ESP8266 Wi-Fi module. From the Cloud, humidity and temperature values can be seen graphically on ThingSpeak platform (<http://thingspeak.com>) from anywhere in the world.

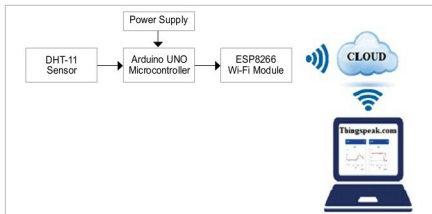


Fig. 1: Block diagram of analysing humidity and temperature using Arduino with ESP8266

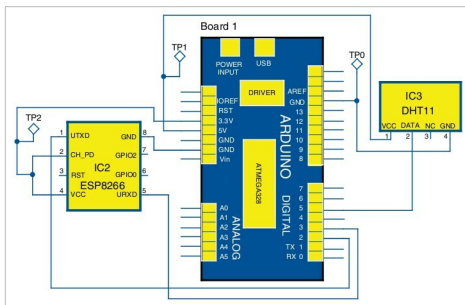


Fig. 2: Circuit diagram for analysing humidity and temperature using Arduino with ESP8266

Construction and testing

ThingSpeak is an open source data platform but you need to register to use it. After registering, login to your account and create a new channel with humidity as one field and temperature as another, as shown in Fig. 3. Once a new channel is created, it will generate two API keys, namely, write API key and read API key. Replace the line given below in the program

Test Points

Test point	Details
TP0	0V (GND)
TP1	5V
TP2	3.3V